THE FOUNDATION v6.0

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Reference Architecture & PoC

Gateway Domain-Centric Routing (GDCR)

A Vendor-Agnostic Metadata-Driven Architecture for Enterprise API Governance

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- 1,499,869 Requests Validated
- 99.9916% Success Rate
- 4 Enterprise Platforms

Navigation

Executive Agenda

01

The Integration Crisis

Three-layer sprawl crisis: proxy, package, and credential proliferation creating structural instability

02

The Solution

Vendor-agnostic metadata-driven routing governance elevating business domains as primary abstraction

03

The DDCR Engine Brain

7-stage deterministic execution flow: Parse → Normalize → Lookup → Route with immutable proxies

04

Universal Business Language

Five languages, one understanding: Japanese, Portuguese, English, Chinese, Italian process unification

05

Cross-Platform Validation

Proven across Kong, SAP BTP, AWS, Azure with 1.5M+ requests and 100% routing success

06

Quantified Impact of the GDCR/DDCR Model

Measurable reductions in complexity, operational effort, and deployment time — validated at enterprise scale.

07

Security Architecture

Semantic URL fakery, fail-fast rejection, whitelist-only routing with defense by design principles

08

Metadata-Driven Scaling Architecture

Infinite vendor onboarding and process expansion — without structural change.

09

DCRP & PDCP: Unified SAP Model

A cohesive SAP BTP implementation where gateway routing (DCRP) and CPI orchestration (PDCP)

10

Architecture as Discipline for Chaos

AI-ready, metadata-driven governance for scalable enterprise integration and future of GDCR

The Integration Crisis

Three-layer sprawl crisis

The Integration Sprawl Crisis - Triple Sprawl

Structural instability from three compounded layers of exponential complexity growth

Proxy Sprawl

1:1 Backend Mapping: Each system exposed through dedicated API proxy

Dozens of Isolated Proxies: Redundant security, rate limiting, monitoring logic

Fragmented Governance: Widened credential surface, increased deployment effort

Traditional Model — 41 Proxies

Package Sprawl

Vendor-Centric Organization: Artifacts named by system, not business domain

Fragmented Packages: Duplicate mappings, inconsistent naming

Weak Domain Visibility: Reduced discoverability, complex lifecycle management

conventions

Traditional Model — 39 Packages

Credential Sprawl

1:1 User Assignment: One technical user per integration artifact

Excessive Proliferation: Complex rotation cycles, increased blast radius

Blackout Risk: Multiple integrations fail simultaneously on misconfiguration

Traditional Model — 39 Users

Sample URLs

/api/v2/sales-orders-create-sf-prod
/api/v1/customer-sync-eu-west-2-s4
/services/v3/finance-invoice-posting-core
/integration/v2/logistics-shipment-dispatch-async

Sample SAP CPI Package names

Salesforce_Orders_Pkg
Shopify_Invoices_Pkg
SAP_ECC_FI_AR_v2_Package
Temp_QA_Customer_Sync_DoNotDelete

Compounded Impact

The impact is not additive — it is multiplicative. Every new backend system results in an additional proxy or, in isolated SAP CPI models, a new package and dedicated technical credentials.

The solution

Gateway Domain-Centric Routing (GDCR)

THE SOLUTION

GDCR: Architectural Overview

Vendor-agnostic framework that treats as the primary architectural abstraction layer

Central Idea

The integration layer must **invert traditional architecture**: technical implementation details move to the background, while the foreground is occupied exclusively by core business semantics.

> **Objects, names, patterns, contracts, and governance structures must speak the language of business domains, entities, and processes — not the language of vendors, protocols, or infrastructure.**

Semantic Unification

APIs mirror business-domain language. External interfaces Organized by business capability, not by system or vendor.
All protocol types resolved dynamically at runtime.

Technical Isolation

Vendor implementations remain invisible to consumers. Backend topology, environment details, and orchestration logic completely abstracted from public APIs.

Standardized Governance

Routing logic stays stable, independent of vendor technology cycles. Immutable proxy artifacts with metadata-driven backend evolution.

7-Step Deterministic Lifecycle

1. **Parse**
   Context extraction from semantic paths
2. **Normalize**
   Canonical action normalization (241→15)
3. **Lookup**
   KVM metadata resolution
4. **Route**
   Dynamic backend binding

Architectural Invariant: Proxy and routing engine remain immutable. Only metadata control plane evolves.

Result: Simplified Surface — Complex inside. Simple outside.

The DDCR Engine

Gateway Domain-Centric Routing (GDCR)

DDCR — Domain-Driven Centric Router

Domain-Driven Centric Router: 7-Stage Deterministic Execution Flow

Deterministic 7-Stage Lifecycle

1 Context Extraction

Extract proxy.pathsuffix, HTTP verb, query parameters, headers. Promote query params to custom headers for consistent interface.

2 Metadata Load

Load domain-specific KVM control plane for current routing context. Metadata acts as system of record.

3 Route Guard Validation

Validate extracted entity against Layer 1 static MatchesPath rules. Unrecognized entities fail fast with HTTP 500.

4 Canonical Normalization

Map **241 action variants** into **15 canonical codes - <4ms**. CREATE→c, READ→r, UPDATE→u, DELETE→d, SYNC→s

5 Routing Key Construction

Generate pattern:

dcrp[entity][action][vendor][iflowid]

Key constructed solely from path segments and canonical code.

6 Metadata Resolution

Resolve KVM for [key]:[adapter]. Positional deterministic schema. No fuzzy matching—exact match or HTTP 500.

7 Dynamic Binding

Establish target URL from KVM value—**never from client input**. Backend URL structurally impossible to manipulate.

Core Mechanisms

Canonical Normalization

Defines valid action space. Without it, Route Guard cannot validate semantically.

Route Guard Validation

Ensures only registered combinations reach metadata resolution.

Dynamic Binding

Safe exclusively because it operates on control-plane-resolved values.

Architectural Truth: Removing any of the three breaks the security and determinism guarantee.

Security Properties

- Whitelist-only routing
- Injection-safe key-value metadata
- Full auditability of metadata changes
- Fail-fast rejection (HTTP 500)
- No path traversal possible

The DDCR Engine: Intelligence Without Complexity

Business intent enters. Metadata decides. Systems remain invisible.

Routing Overhead
<4ms

Semantic Canonicalization Layer

Gateway Domain-Centric Routing (GDCR)

Business Semantic Canonicalization

Multi-Language Business Process Alignment

Process	Japanese	Portuguese	English	Chinese	Italian	GDCR Endpoint
Order → Cash	受注から入金まで	Pedido ao Dinheiro	O2C	从订单到现金	Dall'Ordine al Pagamento	/sales/orders/create
Procure → Pay	購買から支払いまで	Compra ao Pagamento	P2P	从采购到付款	Dall'Acquisto al Pagamento	/procurement/requisitions/approve
Record → Report	記録から報告まで	Registro ao Relatório	R2R	从记录到报告	Dalla Registrazione al Rapporto	/finance/journals/post
Hire → Retire	採用から退職まで	Contratação à Aposentadoria	H2R	从招聘到退休	Dall'Assunzione al Pensionamento	/hr/employees/create
Plan → Produce	計画から生産まで	Planejamento à Produção	P2P	从计划到生产	Dalla Pianificazione alla Produzione	/manufacturing/orders/release
Ship → Deliver	出荷から納品まで	Envio à Entrega	S2D	从发货到交付	Dalla Spedizione alla Consegna	/logistics/shipments/dispatch

Cross-Platform Validation

Gateway Domain-Centric Routing (GDCR)

Cross-Platform Validation

Proven Across Four Enterprise Giants: Kong, SAP BTP, AWS, Azure

Platform Validation Matrix

Kong Gateway — PROVEN
Docker localhost
Implementation: Lua + Redis | Requests: 1,033,600 | Success: 100%

SAP BTP APIM + CPI — PROVEN
Implementation: JS + KVM | Requests: 227,661 | Success: 99.96%

AWS API Gateway + CPI — PROVEN
Implementation: Lambda + Phyton +DynamoDB | Requests: 38,600 | Success: 100%

Microsoft Azure APIM + CPI — PROVEN
Implementation: C# + Redis | Requests: 200,008 | Success: 99.98%

Aggregate Results

Total Requests Processed
1,499,869

End-to-End Success	Routing Success
99.9916%	100%

Failure Analysis
158 Failures = Network-layer only
ECONNRESET / ETIMEDOUT in sandbox environments

Validation Scope

All testing conducted in sandbox/trial environments (SAP aBTP Trial, AWS Free Tier, Azure Trial, Kong Docker localhost). Results demonstrate architectural correctness and portability, not production SLA guarantees.

Test Period
Feb 1-26, 2026

Quantified Impact of the GDCR/DDCR Model

Gateway Domain-Centric Routing (GDCR)

Operational & Performance Impact

Before/After Transformation: Dramatic Efficiency Gains

Transformation Comparison

Metric	Before GDCR	After GDCR	Improvement
API Proxies	41	4	90% ↓
Orchestration Packages	39	4	90% ↓
Technical Users	39	12	69% ↓
Deployment Time	273 min	14.5 min	95% ↓
Vendor Onboarding	Hours/Days	30 seconds	99% ↓
Routing Overhead	—	<5ms	Deterministic
Routing Failures	—	0 / 1.5M	100%
Cross-Atlantic p99	—	338ms	PL→USA
End-to-End Success	—	99.9916%	Validated

Key Achievements

30s
Vendor Onboarding
From hours/days to seconds

<5ms
Routing Overhead
Deterministic latency

0
Routing Failures
Out of 1.5M requests

Security Architecture
Deterministic Security Model

Gateway Domain-Centric

Routing (GDCR)

Deterministic Security Model

Semantic URL Fakery, Fail-Fast Rejection, Whitelist-Only Routing

Semantic URL Fakery

Public Endpoint

/sales/orders/create

Clean, business-semantic, immutable

↓

Backend Resolution (Internal Only)

https://tenant/http/dcrp/orders/c/salesforce/id01

Never exposed to consumers

What Public URLs NEVER Reveal:

- Infrastructure
- topology
- Environment details
- Vendor
- identities
- Orchestration logic

Fail-Fast Rejection

Whitelist-Only Execution

Only metadata-registered route combinations are executable. Unregistered domain/entity/action/vendor combinations return HTTP 500 before any backend contact.

Deterministic Rejection

Invalid combinations rejected before metadata resolution or backend interaction. No probing, no fallback, no fuzzy matching.

Prevent Enumeration

Fail-fast at routing layer prevents lateral exploration and injection attacks. Backend URLs never constructed from client input.

Security Architecture
Authentication Patterns & Integration Modes

Gateway Domain-Centric Routing (GDCR)

Deterministic Security Model

Semantic URL Fakery, Fail-Fast Rejection, Whitelist-Only Routing

Authentication Patterns

OAuth2 /OIDC
User-Facing

Remote token introspection for consent-required, audit-heavy scenarios. High latency due to external dependency.

Basic Auth
Legacy

KVM credential validation for legacy system integration, internal trust zones. Sub-5ms latency.

Header Key
M2M

KVM key lookup for high-throughput machine-to-machine, API key validation. Sub-5ms latency.

Fail-Fast Semantic
GDCR Native

In-memory metadata validation for GDCR native M2M routing. Sub-4ms latency. Control plane acts as authorization boundary.

GDCR Native Security Properties

- Whitelist-only routing
- Fail-fast HTTP 500
- Injection-safe metadata
- No path traversal
- Full auditability
- Zero backend exposure

Security by Design: Semantic URL abstraction is not security through obscurity—it is enforced through deterministic metadata validation and whitelist-only resolution.

Metadata-Driven Scaling Architecture

Gateway Domain-Centric Routing (GDCR)

Deterministic Security Model

Semantic URL Fakery, Fail-Fast Rejection, Whitelist-Only Routing

KVM Metadata Schema: The GDCR Routing DNA

String Format:

[prefix][entity][action][vendor][iflowid]:[adapter]

prefix	entity	action
dcrp | gdcr Framework ID	orders Business entity	c | r | u | d | s Canonical code
vendor	**iflowid**	**adapter**
salesforce Backend system	id01...id99 Flow sequence	http | cxf Protocol type

Two-Level Scaling Strategy

Horizontal Scaling — Vendor Onboarding — Microsoft Dynamics 365

New vendor in existing process = one metadata entry, zero proxy changes

< 60 seconds

Vertical Scaling — Process Expansion — Revenue Recognition

New business process = one static route deployment, then unlimited vendors via metadata

-30 seconds

Implementation Examples

Salesforce Order Creation (REST)

Metadata: dcrporderscsalesforceid01:http Endpoint: /sales/orders/create/salesforce

SAP S/4HANA Invoice (SOAP)

Metadata: dcrpinvoicescs4hanaid10:cxf Endpoint: /finance/invoices/create/s4hana

The Business Intent is Now the Index. Technology is Just the Result.
One proxy. Unlimited vendor scalability via metadata only. Zero operational friction.

DCRP & PDCP: Unified SAP Model

Gateway Domain-Centric Routing (GDCR)

SAP BTP Integration Suite – APIM & CPI as DCRP & PDCP : Unified Model

Domain-centric execution patterns for gateway and orchestration layers

DCRP Pattern

Domain-Centric Routing Pattern

Consolidates multiple vendors and backend systems behind a single, domain-aligned API façade

1. **Organization by Domain**
   APIs grouped by business domain (Sales, Finance, Logistics)
2. **Agility via Metadata**
   KVM updates for routing changes without redeploy
3. **Protocol Agnostic**
   REST and SOAP handled transparently

HYBRID ROUTING MODEL

Static Guards + Dynamic Resolution

Layer 1: MatchesPath rules segment traffic by entity
Layer 2: JavaScript resolves vendor via KVM

PDCP Pattern

Package Domain-Centric Pattern

Reorganizes orchestration artifacts by business domain rather than by system

- One package per domain replaces vendor-centric proliferation
- iFlow DNA naming for deterministic discovery
- Tiered credential governance by business criticality

iFLOW DNA FORMAT

id[seq].[subdomain].[sender].[entity].[action].[receiver].[direction].[sync|async]

Example: id01.o2c.salesforce.order.s4hana.c.in.sync

Transformation

BEFORE (Silo Technical)

- Salesforce_Orders
- Shopify_Invoices
- Workday_Employees
- QuickBooks_Payments

↓

AFTER (Domain)

- <prefix><domain<bprocess>integrations
- sap.sales.o2c.integrations
- nx.logistics.le.integrations
- br.procurement.s2p.integrations

Consolidation Benefits

Packages	39 → 4
Tech Users	39 → 12
Reduction	90%

Architecture as Discipline for Chaos

Gateway Domain-Centric Routing (GDCR)

Architecture as Discipline for Chaos

The Foundation for Next-Generation Enterprise Architecture

The Paradigm Shift

Governance Moves from Artifacts to Domains

Integration landscape reasoned exclusively in business domains and processes rather than technical artifacts. Domain-centric semantics replace vendor-centric sprawl.

Immutable Proxies, Evolving Metadata

The proxy is immutable, metadata evolves, and the domain never lies. Backend changes occur without redeployment, downtime, or architectural rework.

Vendor Interchangeability as Property

Vendor technologies become interchangeable execution details rather than structural dependencies. Cross-platform portability validated across Kong, SAP, AWS, Azure.

Future Directions

Multi-Cloud & Hybrid Ready

Foundation for enterprise architecture spanning multiple gateway platforms and cloud providers with unified domain-centric governance.

AI-Augmented Integration

Portable, vendor-agnostic foundation anchored in business language—ready for AI-driven orchestration and intelligent routing decisions.

Enterprise Scale Expansion

Horizontal and vertical scaling strategies enable infinite vendor onboarding and process expansion without structural changes.

Self-Service Configuration

Web-based configuration interface eliminates manual KVM errors, enabling business teams to manage routing without engineering dependency.

> "There are only two choices — either you govern by semantic domains OR be governed by the complexity of your data."

Based on 1,499,869 validated requests, governing by semantic domains is not just possible, it's proven.

GDCR: This is not a theory — it is a testable, measurable, vendor-neutral executable pattern.

The baseline reference architecture for future GDCR evolution.

Thank you

Gateway Domain-Centric Routing (GDCR)

A deterministic, vendor-neutral integration model.

Intellectual Property & Usage

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> GDCR proves that domain-centric architectures with metadata-driven routing are not only conceptually elegant, but operationally executable at enterprise scale.